

Benjamin Hoover

AI Researcher, Engineer, Student

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Education

Machine Learning PhD Student

Aug 2021 - Present

Georgia Tech, Atlanta, GA

Advisor: Duen Horng (Polo) Chau

Master of Engineering: Electrical and Computer Engineering

May 2018

Duke University, Durham, NC

Bachelor of Science in Engineering: Biomedical Engineering

May 2017

Duke University, Durham, NC

Industry Research Experience

IBM Research

Sep '18 - Present

Visual AI Lab, Cambridge, MA

Mentors: Hendrik Strobelt • Dmitry Krotov

Member of the Visual AI Lab where we use visualization to breathe insight and interactivity into powerful AI systems.

Medtronic Diabetes

Summer '17

Algorithms R&D, Northridge, CA

Mentors: Peter Ajemba • Keith Nogueira

Analyze trends in patient blood glucose levels for the sensor R&D team, developing algorithms that were incorporated into product by the end of the summer.

Academic Research Experience

Georgia Institute of Technology

Aug '21 - present

Polo Club at School of Computational Science and Engineering, Atlanta, GA

Mentor: Polo Chau

Member of the Polo Club of Data Science where we innovate scalable, interactive, and interpretable tools that amplify human's ability to understand and interact with billion-scale data and machine learning models.

International Genetically Engineered Machine (iGEM)

Apr '15 - May '17

Lynch Lab, Durham, NC

Mentors: Eirik Adim Moreb • Michael D. Lynch

Designed experiments to analyze CRISPR-Cas9 binding effects in bacteria, using machine learning to discover patterns in the data.

Awards and Honors

NeurIPS Top Reviewer

Oct 2025

Selected as a top reviewer (top 8% of 24k+ reviewers) for NeurIPS 2025

Spotlight Paper Award

Sep 2025

Epanechnikov DenseAM is a spotlight presentation (top 3% of 21k+ submissions) at NeurIPS 2025

Oral Poster Award

Aug 2025

Dense Associative Memory with Epanechnikov energy is an oral poster (top 5 submissions) at ICCV 2025 workshop on Memory and Vision

Oral Paper Award

Jul 2025

Concept Attention is an oral presentation (top 1% of 12k+ submissions) at ICML 2025

Best Poster Award

Jun 2025

Concept Attention is the Best Poster at the CVPR'25 Workshop on Visual Concepts

Best Poster Award

Oct 2024

Transformer Explainer receives Best Poster at IEEE VIS 2024

Best Poster Award

Mar 2024

Energy Transformer receives 2nd place 🥈 in GATech's annual Graduate Poster Symposium

Best Paper Award, Honorable Mention

Jul 2023

DiffusionDB receives a Best Paper, Honorable Mention Award at ACL 2023

Spotlight Paper Award

Dec 2022

HAMUX is spotlighted (top 5 submissions) at the Deep Learning and Differential Equation Workshop at NeurIPS 2022

Highlighted Reviewer

Apr 2022

ICLR 2022

Best Poster Award

Jul 2020

RXNMapper wins Best Poster award at #IOPPoster 2020

Best Demo Award

Dec 2020

LMdiff is Best Demo at NeurIPS 2020

(Runner Up) Best Demo

Dec 2019

exBERT is runner up for Best Demo at NeurIPS 2019

Dean's List

2013-2017

In the highest one third of engineering class by GPA at Duke

Raul Buelvas Award

Apr 2014

Given to one freshman at Duke University who "best exemplifies honor, optimism, and selflessness" as voted by peers and faculty

Publications

Conference

Dense Associative Memory with Epanechnikov Energy

Benjamin Hoover, Zhaoyang Shi, Krishnakumar Balasubramanian, Dmitry Krotov, Parikshit Ram

Neural Information Processing Systems (NeurIPS). 2025.

  **Spotlight, NeurIPS'25 (top 3%)**

ConceptAttention: Diffusion Transformers Learn Highly Interpretable Features

Alec Helbling, Tuna Han Salih Meral, **Benjamin Hoover**, Pinar Yanardag, Polo Chau
International Conference on Machine Learning (ICML '25). 2025.

[Code](#) [PDF](#) [Video](#) [Demo](#)

🏆 Oral Spotlight, ICML'25 (top 1%) 🏆 Best Paper, VisCon Workshop@CVPR'25

Dense Associative Memory through the Lens of Random Features

Benjamin Hoover, Polo Chau, Hendrik Strobelt, Parikshit Ram, Dmitry Krotov
Neural Information Processing Systems (NeurIPS). 2024.

[Code](#) [PDF](#) [Poster](#) [Video](#)

Transformer Explainer: Interactive Learning of Text-Generative Models

Aeree Cho, Grace C. Kim, Alexander Karpekov, Alec Helbling, Zijie J. Wang, Seongmin Lee, **Benjamin Hoover**, and Others
IEEE Transactions on Visualization and Computer Graphics (VIS). 2024.

[Demo](#) [PDF](#) [Code](#) [Video](#)

🏆 Best Poster @VIS '24 🏆 Went Viral on X (❤️ 1.9k+) 🏆 #1 paper of the day👏

Energy Transformer

Benjamin Hoover, Yuchen Liang, Bao Pham, Rameswar Panda, Hendrik Strobelt, Polo Chau, Mohammed J. Zaki, and Others
Conference on Neural Information Processing Systems (NeurIPS). 2023.

[PDF](#) [Poster](#) [Home](#) [Code](#) [Video \(5m\)](#) [Video \(45m\)](#)

ConceptEvo: Interpreting Concept Evolution in Deep Learning Training

Haekyu Park, Seongmin Lee, **Benjamin Hoover**, Austin Wright, Omar Shaikh, Rahul Duggal, Nilaksh Das, and Others
ACM International Conference on Information and Knowledge Management (CIKM). 2023.

[PDF](#)

Diffusion Explainer: Visual Explanation for Text-to-image Stable Diffusion

Seongmin Lee, **Benjamin Hoover**, Hendrik Strobelt, Zijie J. Wang, ShengYun Peng, Austin Wright, Kevin Li, and Others
IEEE Transactions on Visualization and Computer Graphics (VIS). 2024.

[Demo](#) [PDF](#) [Video](#) [Code](#)

DiffusionDB: A Large-scale Prompt Gallery Dataset for Text-to-Image Generative Models

Zijie J. Wang, Evan Montoya, David Munechika, Haoyang Yang, **Benjamin Hoover**, Polo Chau
Association for Computational Linguistics (ACL). 2023.

[Home](#) [Code](#) [Dataset](#) [PDF](#)

🏆 Best Paper Award, Honorable Mention 🏆 Went Viral 🏆 Oral Presentation

Fairness Evaluation in Text Classification: Machine Learning Practitioner Perspectives of Individual and Group Fairness

Zahra Ashktorab, **Benjamin Hoover**, Mayank Agarwal, Casey Dugan, Werner Geyer, Hao Bang Yang, Mikhail Yurochkin
Conference on Human Factors in Computing Systems (CHI). 2023.

[PDF](#)

Interactive and Visual Prompt Engineering for Ad-hoc Task Adaptation with Large Language Models

Hendrik Strobelt, Albert Webson, Victor Sanh, **Benjamin Hoover**, Johanna Beyer, Hanspeter Pfister, Alexander Rush
IEEE Transactions on Visualization and Computer Graphics (VIS). 2022.

[PDF](#) [Demo](#)

Shared Interest: Measuring Human-AI Alignment to Identify Recurring Patterns in Model Behavior

Angie Boggust, **Benjamin Hoover**, Arvind Satyanarayan, Hendrik Strobelt
ACM Conference on Human Factors in Computing Systems (CHI). 2022.

[Project](#) [Demo](#) [Video](#) [PDF](#) [Code](#)

Can a Fruit Fly Learn Word Embeddings?

Yuchen Liang, Chaitanya K. Ryali, **Benjamin Hoover**, Leopold Grinberg, Saket Navlakha, Mohammed J. Zaki, Dmitry Krotov
International Conference for Learning Representations (ICLR). 2021.

[Project](#) [PDF](#) [Poster](#) [Code](#)

The Design and Development of a Game to Study Backdoor Poisoning Attacks: The Backdoor Game

Zahra Ashktorab, Casey Dugan, James Johnson, Aabhas Sharma, Dustin Ramsey Torres, Ingrid Lange, **Benjamin Hoover**, and Others
International Conference on Intelligent User Interfaces (IUI). 2021.

[PDF](#)

CogMol: Target-Specific and Selective Drug Design for COVID-19 Using Deep Generative Models

Enara Vijil, Payel Das, Samuel Hoffman, Hendrik Strobelt, Inkit Padhi, Kar Wai Lim, **Benjamin Hoover**, and Others
Neural Information Processing Systems (NeurIPS). 2020.

[Project](#) [Demo](#) [PDF](#)

exBERT: A Visual Analysis of Transformer Models

Benjamin Hoover, Hendrik Strobelt, Sebastian Gehrmann
Association for Computational Linguistics: System Demonstrations (ACL). 2020.

[Project](#) [Demo](#) [PDF](#) [Video](#) [Code](#)

Journal

Operational response simulation tool for epidemics within refugee and IDP settlements

Joseph Bullock, Carolina Cuesta-Lazaro, Arnau Quera-Bofarull, Anjali Katta, Katherine Hoffmann Pham, **Benjamin Hoover**, Hendrik Strobelt, and Others
Public Library of Science: Computational Biology (PLOS). 2021.

[PDF](#)

RXNMapper: Unsupervised Attention Guided Atom Mapping

Philippe Schwaller, **Benjamin Hoover**, JL Reymond, Hendrik Strobelt, Teodoro Laino
Science Advances (AAAS). 2020.

[Project](#) [Demo](#) [PDF](#) [Video](#) [Code](#)

Managing the SOS Response for Enhanced CRISPR-Cas-Based Recombineering in E. coli through Transient Inhibition of Host RecA Activity

Eirik Adim Moreb, **Benjamin Hoover**, Adam Yaseen, Nisakorn Valyasevi, Zoe Roecker, Romel Menacho-Melgar, Michael D. Lynch
American Chemical Society: Synthetic Biology (ACS). 2017.

[PDF](#)

Workshop

Memory in Plain Sight: A Survey of the Uncanny Resemblances between Diffusion Models and Associative Memories

Benjamin Hoover, Hendrik Strobelt, Dmitry Krotov, Judy Hoffman, Zsolt Kira, Polo Chau
NeurIPS Workshop on Associative Memory and Hopfield Networks (AMHN@NeurIPS). 2023.

[PDF](#) [Poster](#) [Home](#)

HAMUX: A Universal Abstraction for Hierarchical Hopfield Networks

Benjamin Hoover, Polo Chau, Hendrik Strobelt, Dmitry Krotov
Symbiosis of Deep Learning and Differential Equations II Workshop at NeurIPS (DLDE). 2022.

[Code](#) [Project](#) [PDF](#) [Video](#)

Preprints

Tutorial on Modern Methods in Associative Memory

Dmitry Krotov, **Benjamin Hoover**, Parikshit Ram, Bao Pham
International Conference on Machine Learning (ICML '25). 2025.

[Code](#) [PDF](#) [Home](#)

Fairytailor: A multimodal generative framework for storytelling

Eden Bensaid, Mauro Martino, **Benjamin Hoover**, Hendrik Strobelt
ArXiv preprint (ArXiv). 2021.

[PDF](#) [Demo](#)

Talks

Emergent Memories in Dense Associative Memories

Memory & Vision Workshop @ ICCV 2025 [Invited](#)

Oct 2025

Panelist: Integrating Memory & Vision

Memory & Vision Workshop @ ICCV 2025 [Invited](#)

Oct 2025

A New Era for Associative Memories

Georgia Tech-IBM TechXchange [Invited](#)

Aug 2025

Modern Methods in Associative Memory

ICML 2025 Tutorial

July 2025

ConceptAttention: Visualize Any Concept in Your Generated Images

CVPR Workshop on Visual Concepts [Invited](#)

June 2025

Panel Moderator: Improving LLM's with Memory

New Frontiers in Associative Memory Workshop @ ICLR 2026 [Invited](#)

Apr 2025

Hopfield Networks 2.0: Associative Memory for the Modern Era of AI

Physics Informed ML Workshop at Los Alamos National Laboratory [Invited](#)

Sep 2024

Plectics Lab Colloquium [Invited](#)

May 2024

Georgia Tech CSE Hotseat

Nov 2022

Panelist: Future of Software Engineering in Associative Memory

Associative Memory & Hopfield Networks Workshop @ NeurIPS 2023 [Invited](#)

Dec 2023

Energy Transformer

McMahon Lab at Cornell University [Invited](#)

Sep 2023

Comparing Language Models with LMDiff

Demo at NeurIPS 2020

Dec 2020

exBERT: Exploring Learned Embeddings in Transformer Models

System Demonstration at ACL 2020

Jul 2020

IBM Exposition Demo at ICLR 2020

Apr 2020

System Demonstration at NeurIPS 2019

Dec 2019

Unsupervised Attention Guided Atom Mapping with RXNMapper

ML Interpretability for Scientific Discovery Workshop at ICML 2020

Jul 2020

IBM Exposition Demo at ICML 2020

Jul 2020

Breaking the Wall of Black Box AI

Falling Walls Lab Boston

Oct 2020

The Inside Story

MEng Graduation elected speaker Invited

May 2018

Using Matlab's PRT to Accelerate Algorithm Development

Medtronic Diabetes R&D

Jul 2017

The Impact of Encouragement

Cary Christian School Valedictorian Speech Invited

May 2013

Teaching

Graduate Teaching Assistant

Fall '23 - Spring '24

Georgia Institute of Technology, Atlanta, GA

Instructors: Polo Chau • Mahdi Roozbahani

Organized project teams, graded projects, and held weekly office hours for an online graduate course with 947 students enrolled.

Graduate Teaching Assistant

Spring '18

Duke University, Durham, NC

Instructor: Patrick Wang

Designed homeworks, supervised projects, and occasionally teach lectures.

Graduate Teaching Assistant

Spring '18

Duke University, Durham, NC

Instructor: Jonathan Viventi

Supervised and graded capstone projects where students designed and assembled 2-channel EEG readers

Graduate Teaching Assistant

Spring '18

Duke University, Durham, NC

Instructors: Stacy Tantum • Paul Bendich

As a TA, I helped setup class infrastructure (Python, jupyter) and initial curriculum for this 25 student pilot course for Duke's Data+ Program.

Undergraduate Teaching Assistant

Spring '17

Duke University, Durham, NC

Instructor: Jonathan Viventi

Prepared, supervised, and graded labs where students designed medical devices such as Ultrasound, ECGs, Baby Incubators, and functional FitBits.

Undergraduate Teaching Assistant

Jan '15 - May '18

Duke University, Durham, NC

Instructor: Stacy Tantum

Graded homeworks, held weekly office hours, supervised labs, and private tutored students.

Boeing Fellow

Jan '16 - May '17

Duke University, Durham, NC

Instructor: Carmen Rawls

Wrote STEM lesson plans to engage middle and high school students in the Durham area.

Service

Reviewer

IEEE Transactions on Network Science and Engineering (**TNSE**) 2025
Neural Information Processing Systems (**NeurIPS**) 2025 · 2024 · 2023 · 2021
International Conference for Learning Representations (**ICLR**) 2025 · 2024 · 2023 · 2022
International Conference for Machine Learning (**ICML**) 2025 · 2023 · 2022
ACM Conference on Human Factors in Computing Systems (**CHI**) 2023
Workshop on Memory and Vision at ICCV (**MemVis@ICCV**) 2025
Workshop on New Frontiers in Associative Memories at ICLR (**NFAM@ICLR**) 2025
Workshop on Associative Memory & Hopfield Networks at NeurIPS (**AMHN@NeurIPS**) 2023
Workshop on the Symbiosis of Deep Learning and Differential Equations at NeurIPS (**DLDE@NeurIPS**) 2022
Blogpost Track at ICLR (**BP@ICLR**) 2023

Organizing Committee

Workshop on Associative Memory & Hopfield Network at NeurIPS (**AMHN@NeurIPS**) 2023
Workshop on New Frontiers in Associative Memory at ICLR (**NFAM@ICLR**) 2025

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